

PREDICTING STOCK MARKET PRICES WITH FUNDAMENTAL INDICATORS AND TECHNICAL INDICATORS

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The purpose of this research paper is to use machine learning to forecast future stock prices and trends using both fundamental analysis and technical analysis indicators as explanatory features of the model. Fundamental analysis and technical analysis are two of the most widely used investment analysis techniques. Fundamental analysis attempts to measure a stock's intrinsic value, whereas technical analysis uses historical prices and charts to determine the true value of a stock. Both these strategies are often seen as competing because they differ in how a stock's value is determined. However despite these differences, this paper attempts to combine these strategies as a hybrid strategy to predict future stock prices.

To do this, a comparison is made of three neural networks; Model 1: Technical Model which includes a neural network with technical indicators. Model 2: Fundamental Model with fundamental indicators. Lastly, Model 3: The Hybrid Model, which includes both fundamental and technical indicators as inputs to the model.

The results show that the hybrid model with both fundamental indicators and technical indicators is the best performing model based on the SSE computed.

Key words:

1. Introduction

The stock market has been a fascinating topic which has gained immense popularity since its development in the 16th century. It has grown and evolved from one company being listed to over 40 000 listed companies all over the world. By definition, the stock market is the collection of exchanges and platforms where shares of public companies are bought and sold for profit. Subsequently, from a company's perspective, it is one of the most important ways for a company to raise capital. The main goal for any investor is to find the most valuable companies on the stock market which will guarantee consistent returns in the future. Hence, stock prediction is important for any investor because accurate determination of the future value of a stock could give rise to greater returns.

The ability to be able to look into the future is a skill most investors want to possess, but because of the nature and complexity of the stock market, prediction has proved to be a challenging task and a fascinating topic of interest Agrawal, Chourasia and Mittra (2013). Stock price prediction is a challenging task because of the various factors that influence the stock market. Yet, at any point in time, the stock price will always be determined by supply and demand Harper (2021). Nonetheless, traders and investors have been successful in beating the market and realising abnormal profits by using various prediction tools and techniques Latif, Arshad, Fatima and Farooq (2011).

The two most prevalent stock analysis techniques that most investors rely on are Fundamental analysis and Technical Analysis. The former approach attempts to determine a company's intrinsic value by analysing a companies reports and external factors that can affect normal business practises Drakopoulou (2016). On the contrary, Technical Analysis involves modeling historical prices to determine future trends. Technical analysts do not measure a stocks intrinsic value, instead they analyse trends, patterns and other factors affecting a stocks price. Masry et al. (2017). The limitations of traditional techniques include prediction inaccuracies caused by incorrect use of models, incorrect interpretations. This is because most traditional forecasting models depend heavily on human expertise which can be very subjective Patalay and Bandlamudi (2020). It is for these reasons which has led investors and researchers to shift from traditional forecasting methods to machine learning tools.

Machine learning is the field of study which enables computer software the ability to extract knowledge from data without being programmed to do so Mahesh (2020). Machine learning based techniques have gained a lot attention in the development of trading systems due to their capacity to learn dependency patterns Lara-Benítez, Carranza-García and Riquelme (2021). Deep learning algorithms form as a subset of Machine learning and have introduced new methods to predictions by modelling the relationships between variables in deep multi-layers Namin and Namin (2018). The most commonly used deep learning algorithms include auto-encoder neural networks, deep belief networks and recurrent neural networks. Several types of recurrent neural networks are used in forecasting financial time series where its main function is to process and store sequential data as memory for predictions Ma (2020). However, the limitations with recurrent neural networks is its diminishing returns formally known as the vanishing degrading problem. This involves an exponential decrease in the gradient such that the algorithm learns at a very slow rate. This problem and has been solved by the development of the Long and short-term memory algorithm Nikou, Mansourfar and Bagherzadeh (2019).

The main function of Long and short term memory is to solve the vanishing gradient problem and to analyse longer time series data Ma (2020); ?. The special feature of LSTM approach is its ability to store historical data for longer time periods using three special cells namely the input gate, output gate and the forget gate. Di Persio and Honchar (2017). Therefore, the use of LSTM models are more appropriate for the prediction of timed data as they are able to handle long sequences of input data compared to other neural network approaches Nelson, Pereira and De Oliveira (2017).

The main objective of this research is to show that fundamental analysis and Technical analysis can be used as a hybrid strategy to forecast future prices and trends. Therefore the goal is to build a deep learning algorithm with both fundamental indicators and technical indicators to forecast next months closing stock price.

Research objectives

- To explore fundamental analysis and technical analysis
- To Determine and select fundamental indicators and technical indicators that will be used as features in the model
- To Build a long short-term memory algorithm as the model of choice to predict future stock prices and trends using different indicators

- Evaluate the different models using statistical methods to determine whether using fundamental indicators and technical indicators as a hybrid strategy could lead to a better performing model in terms of prediction accuracy.

2. Theoretical Background

2.1 Fundamental Analysis

Fundamental Analysis is a method or technique used by financial practitioners and participants of the stock market to determine the intrinsic value of a stock, that is, the current worth of a security or stock. It involves examining financial statements and macroeconomic factors to determine whether the true value of a company is reflected in its price and whether an investment is worth the risk Drakopoulou (2016).

The main goal for any investor is to invest in a company that will maximise their wealth. When a company is performing well this creates opportunities for higher returns because there is a higher possibility of a rise in prices. Therefore, in the long-term, good fundamentals are always reflected in the stock prices Luckieta, Amran and Alamsyah (2020). The starting point for any fundamental analyst involves examining financial statements to understand the financial position and performance of a company. Analysing financial statements is a difficult task and the numbers disclosed on statements alone are not very useful to investors. To be able to assess the financial position of a company, fundamental analysts use financial ratios to measure the relationship between financial variables that appear on financial statements. Heikal, Khaddafi and Ummah (2014). There are several different types of financial ratios that can be used to analyse financial statements. These ratios can be divided into categories which explore different aspects of a company's financials and performance. In general, There are five main categories that financial ratios can be broken down into such as measures of liquidity, profitability, debt, operating and investment valuation Arkan (2016).

- **Valuation financial ratios:** Valuation ratios are ratios used to determine the market value of a stock based on the relationship between stock price and company fundamentals. Examples of valuation ratios include Price/ Earning , Price/Book value and Earnings per share Arkan (2016). These ratios measure how much investors are willing to pay more than what the company is worth, thus these ratios depend on the current stock price and will change as the price changes Herawati, Putra et al. (2018).
- **Liquidity financial ratios:** This group of financial ratios is focused on understanding how well the company manages its short-term debt Krantz (2016). These ratios measure the relationship between current assets and current debt and determine how effective a is at meeting its short-term obligations. Examples of liquidity ratios include Current Ratio, Debt-to-Equity ratio and the acid test ratio Arkan (2016)
- **Leverage Financial Ratios:** Debt ratios are commonly used to measure the proportion of debt relative to equity used to finance the operations of a company Heikal et al. (2014). Therefore, the higher the debt obligation, that is, the company is more dependent on debt financing than equity financing, the higher the risk. Examples of Debt ratios include Debt ratio, Debt to Equity and short-term debt to equity Luckieta et al. (2020).

- **Financial Operations ratios:** Generally investors are interested in how well a company can organise its resources to generate a profit, since an increase in profit means more opportunities to generate wealth for investors Krantz (2016). Therefore the role of Operating performance ratios is to measure how well a company can utilise its financial resources in order to increase sale. Examples of performance ratios include Total Asset turnover and Fixed asset turnover Herawati et al. (2018).

Therefore, by using financial ratios to study financial statements investors are able to compare company financials of competing companies, to determine the risk associated with an investment, and make well-informed trading decisions during market fluctuations Krantz (2016)

Fundamental analysis goes beyond studying the financials of a company ,it also involves examining macroeconomic factors that can impact the performance of a company and its stock price Drakopoulou (2016). The relationship between macroeconomic factors and the stock market is based on the fact that the stock market creates opportunities for cheaper funding for businesses and government institutions and as a result businesses expand, more jobs are created, people have more money to spend, and the overall economy benefits from this interaction Jareno Cebrian and Negrut (2016). Therefore, the stock market has an impact on macroeconomic factors and the reverse is also true, because sudden changes in macroeconomic activities can lead to uncertainty in future gains which will affect stock returns Ahmad and Ramzan (2016). Therefore, it is important for any investors to consider this relationship when making investment decisions. Macroeconomic factors are events such as fiscal, natural or geopolitical events that have the potential to affect a population. Examples of macroeconomic factors include Inflation, exchange rates, money supply and Gross domestic product to mention the least Ndlovu, Faisa, Resatoglu and Tursoy (2018). Many economists and analysts have studied whether there exists a relationship between the real economy and the stock market and evidence suggests that there exists a relationship, however, there still remains uncertainty on whether this relationship is positive or negative Jareno Cebrian and Negrut (2016). Different factors each affect the stock market in different ways, for example, there exists a negative relationship between interest rates and the stock market, and in contrast a positive relationship between GDP and the stock market Ndlovu et al. (2018). When interest rates increase, businesses accumulate expenses as the cost of lending has increased, this results in a decrease in profits. As a consequence, the stock market will shift into a bear market which is a gradual reduction in the value of stocks Jareno Cebrian and Negrut (2016).

2.2 Technical Analysis

Technical analysis is an investment approach that uses a range of trading strategies that analyse historical prices to predict future stock price trends. This approach is based on the idea that future stock prices are anticipated by historical prices and that history tends to repeat itself Masry et al. (2017). Technical analysis attempts to understand the market sentiments behind price trends by looking for patterns and trends rather than analysing a security's fundamental attributes. Technical analysis offers a different perspective to analysing stock prices because it develops graphs and charts based on the history of trading the volume and price changes. The objective for any technical analyst is to determine the time of direction change, and whether it will be a downward trend or upward trend in order to make a decision of whether it is suitable to enter or exit the market. Masry et al. (2017)

Technical analysts use stock price data such as; opening price, closing price, and volume, in formulae to produce a series of data points. The generated series is known as a Technical indicator and can be plotted against a price chart to form as a basis for comparison. Fernández-Blanco, Bodas-Sagi, Soltero and Hidalgo (2008). There are various technical indicators that are used which differ according to the complexity of the formula and its implementation. Technical indicators can be grouped into two broad classes; namely oscillators and lagging indicators Fernández-Blanco et al. (2008). Stochastic oscillators such as the KD Line and Relative strength index are leading indicators as they have the ability to signal a change in price trend before it has occurred. A signal occurs when the closing price approaches the lowest price in that period during a bullish trend, this could indicate a move towards a downward trend. On the other hand, when there is a bearish trend, oscillators will measure when the closing price will approach the highest price in that period, which will indicate the end of a bearish trend and move towards a bullish trend. Nazario, e Silva, Sobreiro and Kimura (2017). Trend following indicators or lagging indicators are indicators that follow price action and tend to only become apparent after a large shift has taken place. These indicators are commonly used to confirm long-term trends rather than to predict future trends. Examples of lagging indicators include Moving Average (MA), Exponential moving average (EMA) and Moving Average Convergence-Divergence (MACD) Fernández-Blanco et al. (2008).

2.3 Machine learning

Machine learning is a branch of artificial intelligence that was developed to give computers the ability to learn without being programmed to do so. Machine learning algorithms are able to automatically interpret and learn from complex data to make accurate predictions. With the massive amount of computational ability, machine learning algorithms can solve problems at a speed and scale that cannot be duplicated by human expertise. Generally, machine learning algorithms simulate human learning and use historical data to predict new output values Nikou et al. (2019). The type of algorithm used to solve a problem depends on many factors such as the data used, the number of variables and the kind of model used to predict the outcomes Mahesh (2020). Machine learning algorithms have made a big impact in the field of finance and especially in stock market prediction. Predicting stock market prices has been a challenging task for decades and there has been a lot of controversy with regards to the true nature of the stock market. Many economists and researchers in this field believe that stock prices follow a random walk process, whereas others such as technical analysts have shown evidence that stock prices are not random but in fact follow a non-linear pattern based on the past Choudhry and Garg (2008). Since the stock market is affected by many different factors, its complex non-linear nature cannot be fully captured effectively by traditional forecasting models. However, in the past decade machine learning algorithms have proven to be the best tool used to accurately predict stock price movement due to its ability to learn the complex relationship between nonlinear variables Adebisi, Ayo, Adebisi and Otokiti (2012). The most frequently used machine learning algorithms implemented in the finance industry are Artificial neural networks, Support Vector Machines and Ensemble learning Macchiarulo (2018)

Artificial Neural networks (ANN)

Artificial Neural Networks or neural nets are machine learning algorithms that were developed to mimic the way in which the human brain processes information. According to Nikou et al. (2019) Artificial neural networks have been the most important non-linear models in financial markets that

have achieved the desired results in predictions..ANN's fall into the category of prediction models where results are obtained by learning patterns from original data and make inferences about the outcome Ma (2020).Therefore, neural nets the adaptability and flexibility to handle data that fluctuates in short periods of time, which makes it a perfect candidate for stock price prediction Adebisi et al. (2012). ANN's are based on the biological neural system, and similar to the human brain they are built with neurons (or nodes) which process information and connections that pass the information from one neuron to the next Kurani, Doshi, Vakharia and Shah (2021). A typical Neural Network can be seen in (Figure.1), which has neurons that are connected to other neurons in various layers of the network As illustrated in figure 1, a neural network has three layers; an input layer, a hid-

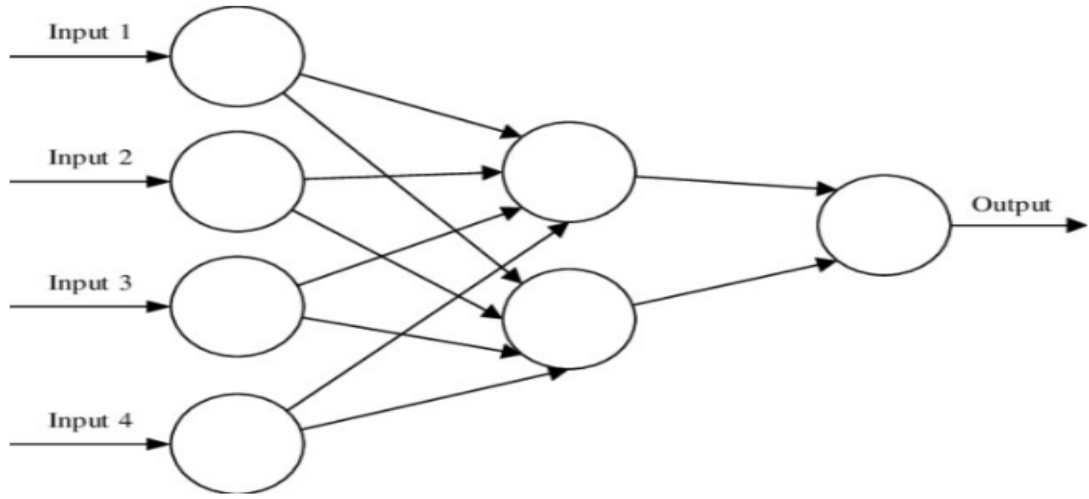


Figure 1. Figure 1:Artificial Neural Network

den layer and an output layer. Information is fed into the neural network via the input layer which takes values corresponding to independent variables or features of a model.The number of features in a neural network determines the number of nodes in the input layer Siami-Namini and Namin (2018).The input values are then passed on to the next layer via the connections. Each connection is parameterized with a weight to illustrate the strength of a connection. Therefore, the inputs to the hidden layer are a weighted sum of the input values from the input layer and the weights from each connection. In the hidden layer, the weighted sum of input values are processed within each neuron by an activation function. Lastly, the output values from the hidden layers are passed forward to the output layer where the values are processed once again to produce a predicted value Ganatr and Kosta (2010). This process is known as a feedforward pass and is expressed mathematically as;

Input: x_1 and x_2
Weights: $w_{11}^1, w_{12}^1, w_{21}^1, w_{22}^1$
Hidden layer neurons: H_1 and H_2
bias terms: b_1 and b_2
activation function: σ_1 and σ_2

output layer neuron: O_1

Input layer:

$$\begin{aligned}x_1 &\rightarrow x_1 * w_{11}^1 \\x_2 &\rightarrow x_2 * w_{12}^1\end{aligned}$$

Hidden layer:

$$\begin{aligned}H_1 &= w_{11}^1 x_1 + w_{21}^1 x_2 + b_1 \\H_2 &= w_{12}^1 x_1 + w_{22}^1 x_2 + b_2\end{aligned}$$

Activation functions:

$$\begin{aligned}\sigma_1 &= \sigma_1(H_1) \\ \sigma_2 &= \sigma_2(H_2)\end{aligned}$$

Output layer:

$$O_1 = w_{11}^2 \sigma_1 + w_{21}^2 \sigma_2 + b_3$$

2.4 Activation Functions

Activation functions are mathematical functions that are used in neural networks to transform input values (weighted sum of inputs) into an output. Activation functions play a big role in the neural network training process and thus the choice of an activation function can impact the performance of the network Gamboa (2017). There are many activation functions that can be used in neural networks but the most commonly used include the sigmoid function, the rectified linear function and the hyperbolic tangent function.

1. **Sigmoid/logistic function-** The sigmoid function or logistic function is an example of a non-linear activation function. It can take any real value as an input and the output is a value between 0 and 1. The function is defined as:

$$\sigma = \frac{1}{1+e^{-x}}$$

The larger the input value the closer the output value will be to 1, and the smaller the input value or more negative, the closer to zero. The logistic function is commonly used in situations where the objective of the network is to predict a probability Sharma, Sharma and Athaiya (2017)

2. **Rectified linear activation function (ReLU)-** The ReLU function is a piecewise function that outputs the input value when the input value is positive, or otherwise zero if the input is negative. The ReLU function is calculated as:

$$f(x) = \max(0, x)$$

Unlike other commonly used activation functions, the ReLu function is less vulnerable to the vanishing gradient problem and therefore it is easier to train and performs better Sharma et al. (2017) .

3. **Hyperbolic Tangent**- is sometimes referred to as the tanh function and it is commonly used as an alternative function to the sigmoid. The tanh function can take on any real value that will be transformed as an output value between -1 and 1. The larger and more positive the input value then closer it will be to 1, and the smaller and more negative the input value the closer it will be to -1. The tanh function is calculated as follows:

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

The tanh function is mainly used in networks whose goal is classification Sharma et al. (2017).

The prediction accuracy of a neural network is dependent on the number of layers used and on the type of activation function. In a multilayer perceptron , the same activation function is used in the hidden layer and on occasion a different activation is used in the output layer. There is no specific method or technique that suggests the type of activation function to be used or the number of layers to be added in the neural network, however previous research and studies have show that a minimum of 2 layers is sufficient to reduce error in predictions Sharma et al. (2017).

3. Literature review

Machine learning algorithms have been used to solve many problems in industries such as banking, medicine and the automotive industry to mention the least. For the purpose of this research the main focus will be on the use of machine learning algorithms to predict stock market prices. In this section different studies are reviewed on the use of machine learning algorithms to predict stock market prices and the applications of fundamental analysis and technical analysis to forecast future trends and prices.

Technical analysts generally depend on chart patterns, technical indicators and price fluctuations to make investment decisions. This method of analysis is demonstrated in the following study conducted by Priya and Arabinda (2019) where stock prices of three companies are analysed using descriptive statistics and technical indicators. The opening and closing price fluctuations are analysed using the simple moving averages model. Furthermore, the money flow index and the relative strength index were applied to extract information on the price movement and lastly the beta index was used to study the risk and volatility of the stock relative to the market. The results show that the simple moving averages method gives opportunities for identifying buy and sell signals. Signals are seen when the average is below or above the trailing price. On the other hand, the money flow index indicates whether the stock has been oversold or overbought. lastly, the beta index indicates the level of risk and volatility associated with a stock relative to the market. This study illustrates how technical analysis can be used to identify buy and sell signals and to determine the relationship between the stock and the market.

On the contrary, the following study Herawati et al. (2018) explores the effects of fundamental analysis indicators on stock prices of companies in the food and beverage industry. The study compared the effect of Debt-to-equity ratio (DER) , Return on assets (ROA) , Current ratio (CR) , Price earning ratio (PE) , and Total Asset turnover (TATO) .The results show that ROA And PE ratio have a postive influence on the stock price. Whereas, TATO and CR show no influence on the stock price. Lastly, Debt to Equity shows no negative effect on price. Therefore,It is evident that there exists a relationship between stock prices and financial ratios calculated from company financial statements. To introduce this idea of using fundamental analysis and technical analysis as a hybrid strategy, the following research Beyaz, Tekiner, Zeng and Keane (2018) explores whether the combined use of fundamental analysis and technical analysis is beneficial in stock price forecasting. The results show that forecasting models using fundamental indicators outperform models that use technical indicators specifically for companies in the information technology industry. Furthermore, forecasting models that used both fundamental and technical indicators resulted in a lower RMSE , thus yielding better forecasting performance.This idea of using technical analysis and fundamental analysis as a hybrid strategy is further justified by the following paper Bonga (2015) where the author highlights that by considering both strategies in decision making this makes for an effective investment strategy. Furthermore, by combining both techniques, and ensuring that both sets of indicators agree on the price movement,investors have the potential to realise greater returns.

Traditional time series forecasting methods have always been the first choice when analysing stock market trends. Traditional models such as the autoregressive moving averages model and the exponential smoothing model have shown their superiority by producing accurate results, and offering a simple and robust solution to prediction. However, these methods are limited in their abilities as they heavily depend on human judgement, and can be time-consuming and costly to implement. With the advancement in technology and processing power, machine learning algorithms have shown superior performance over traditional methods Sagheer and Kotb (2019). This topic is further explored in the following article Ma (2020) which compares the predictive accuracy of an ARIMA model, Aritificial neural network and Recurrent neural networks for the prediction of DELL's stock price. The study first compares the prediction results of an ARIMA(1,0,0) and an ANN(10, 17,1). The results show that there is no significant difference in prediction accuracy between the two models. Furthermore, based on the observed results it is evident that the ANN predicts the numerical value of the stock price whereas the ARIMA model predicts the direction of the stock price. Moreover, since ANN's are non-linear in nature, the model was able to capture the fluctuations of the test data better than the ARIMA model. Therefore, the ANN model is seen as the superior model, however, this conclusion may vary for different datasets. The study then compares the prediction results of the ANN and LSTM model. The results conclude that due to the addition of indicators and the improvement of the vanishing degrading problem, the LSTM model is more appropriate and performs better than the ANN. In the following paper Lara-Benítez et al. (2021)of the most recent deep learning algorithms for time series forecasting is conducted and experimentation of different deep learning architectures is . In addition,an experimental study is conducted on seven different deep learning architectures to compare accuracy and efficiency. The study evaluated the different algorithms under different configurations and training parameters.Three categories were reviewed in the study namely the fully connected neural networks, recurrent neural networks and convolutional neural networks. The review results show that recurrent neural networks specifically the Elmans recurrent neural net-

work outperforms logistic regression models and multi-layer perceptrons. However, the algorithm has two major limitations, including the vanish degrading problem and the exploding degrading problem which are later solved by the Long and short-term memory algorithm. Thus, making the latter algorithm the preferred choice for prediction and forecasting

In the recent years Deep learning algorithms such as the Long short-term memory model and convolutional neural network have been known to be the superior choice for sequential data. This point is highlighted and demonstrated in the following study Gamboa (2017) where several deep learning algorithms are reviewed and compared. Furthermore, a common misconception is highlighted in that deep networks yield better results than shallow networks. However, this is not usually the case, Instead the opposite is seen to be true, that shallow networks tend to yield better results. The study further explores different applications of deep learning algorithms such as the task of Classification using Convolutional Neural Networks (CNNs) and Forecasting using several deep learning techniques such as Deep Belief Networks and Stacked Autoencoders which are used for traffic flow predictions. Other less researched Applications of deep learning algorithms include Anomaly Detection of time -series data.

4. Methodology

The goal of this research is to explore the idea that technical analysis and fundamental analysis can be used as a hybrid strategy to forecast future stock prices

Research objectives

- Explore fundamental analysis and technical analysis
- Determine and select fundamental indicators and technical indicators for the model
- Build a long short-term algorithm as the model of choice to predict future prices using different indicators
- Evaluate the different models using statistical methods to determine whether combining fundamental indicators and technical indicators as a hybrid strategy could lead to a better performing model in terms of prediction accuracy

4.1 Data Description

The data used for this research contains stock price data for Google which is listed on the New York Stock Exchange. The dataset was downloaded from Yahoo Finance and Borsdata. Yahoo finance is a financial platform that provides financial news, data, stock quotes, press releases, financial reports and original content. The data retrieved from Yahoo finance includes; Opening stock price, Closing stock price, Adjusted closing price and Volume of transactions. The time period of the dataset is from 2004/08/19 to 2022/04/01.

The following features were selected based on a review of previous research and from online web searchers of the most commonly used financial ratios and technical indicators.

4.2 Technical Indicators

Opening price The opening price is the stock price at the start of the trading day.

Closing price The closing price is the stock price at the end of the trading day. It is the price after the last transaction.

Volume -Volume measures the number of transactions that occur per day in the stock market relative to each stock. When there is high volume of transactions with respect to an increase in stocks price, this indicates a healthy and strong stock. Contrary to this, when there is high volume with respect to a decrease in stock price, this shows a strong trend downwards.

Adjusted closing price This is the closing price of a stock adjusted for corporate action such as stock splits, dividends, right offerings. The adjusted closing price is used to analyse historical returns and to perform detailed analysis as it reflects the stock's value after accounting for corporate action

Moving average : Moving averages are the most commonly used technical indicators. This is because they are the easiest to implement and they enable traders to view stock prices from a less volatile perspective, that is, moving averages smooth out the data to make it easier to interpret the trend Drakopoulou (2016). There are atleast 6 different moving averages that most investors use, but for this research the focus will be on only two of them namely; the simply moving average and the exponential moving average.

The simple moving average (SMA)

is the simplest MA which can be calculated by adding up the most recent stock prices and dividing the total by the number of time periods;

$$SMA = \frac{\sum \text{closing stock prices for the last } N \text{ days}}{N \text{ days}}$$

where N is the number of time periods . The number of time periods used can play a role in how effective the model is. This is because shorter time frames will react to price changes more quicker than a longer time frames. For example, A 20-day moving average will approximate the actual price more closely than a 100-day moving average does. therefore, the 20-day MA will be more beneficial to use. However, this choice is also determined by the goal of the investor. In this research we will determine the best time period to use by comparing the MSE and MAD of the 50-day MA, 100-day and 200-day MA. Thus, the MA period with the smallest error will be chosen as an indicator for the model.

Exponential moving average (EMA)

The exponential moving average (EMA) can be viewed as an improved version of the Simple moving average because it tends to react faster to price changes than the latter. The difference between the two indicators is in the calculation, where the exponential moving average is calculated by taking the weighted average stock price. This technique of calculating the moving average assigned more weight to the most recent prices Drakopoulou (2016). To calculate the EMA, the following formular is used:

$$EMA = (P * a) + PreviousEMA * (1 - a)$$

Where P = current price, a = smoothing factor = $2/(1 + N)$ and N = number of time periods. Similar to the simple moving average, the time frame chosen will determine the effectiveness of the indicator. Therefore, to determine the best time period for the indicator, the same method that used for the SMA will be applied here, where a comparison the errors will made between the 50-day EMA, 100-day EMA and a 200-day EMA.

- **Relative strength Index(RSI):** The RSI is one of a few indicators that fall in the category of momentum oscillators. In general, momentum oscillators are used to measure the magnitude of change in a securities price at given time period Nazario et al. (2017). The RSI is one such example and is calculated as:

$$RSI = 100 - \frac{100}{1 + \frac{(PreviousAverageGain * N) + CurrentGain}{(PreviousAverageLoss * N) + CurrentLosses}}$$

where N denotes the standard number of time periods, which is usually 14 days. As shown in the formular above, the RSI compares the strength of price increases versus the strength of price decreases. Furthermore, it is plotted beneath a stocks price as a line graph that fluctuates between 0 and 100. An RSI reading above 70 indicates an overbought condition, whereas a reading of 30 or below indicates an oversold condition. Thus, investors are able to determine overbought and oversold stocks, to signal price reversals and to validate current trends. Therefore, to use this oscillators as an indicator in the model, a 14-day RSI will be calculated using the above formular on the closing prices Gumparthi (2017).

4.3 Fundamental Indicators

- **Price/Earnings ratio (P-E):** The P-E ratio relates a companies share price to its earnings per share. This ratio measures what the market is willing to pay now for future returns of a company's stock. It is one of the most commonly used ratios that investors use to compare companies in the same industry Krantz (2016).
- **Price/Book value ratio-** This ratio compares the current stock price to the companies book value, which is the value of everything the company owns, free and clear of debt. This ratio is calculated as:

$$P/B = \frac{Stockprice}{Totalassets - Totalliabilities}$$

A strategy employed by fundamental analysts is to invest in a company with a P/B ratio less than the market with the expectation that in the future the market will rectify itself and the price will be corrected to reflect the true value of the stock Drakopoulou (2016)

- **Return on equity-** Return on equity ratio is a performance ratio used to measure a compa-nys ability to generate revenue from sales and assets Purnamasari (2015). This ratio gives an investor a good indications of how well a company can employ invested capital to generate a profit Kijewska (2016). To calculate the ROE:

$$\frac{NetIncome}{Average\text{shareholder's}\text{equity}}$$

Krantz (2016)

- **Debt-to-Equity ratio:** Debt-to-equity ratio is a leverage financial ratio that measures the proportion of debt that a company has taken to finance its operations compared to its equity. It is an important ratio because it highlights a company's dependence on borrowed funds and it focuses on a company's ability to meet long-term debt commitments. A high debt-to-equity ratio means a greater obligation to its lenders which means increased interest expenses. This is an indication of a high risk investment because the company may find it difficult to find external funding when its current debt-to-equity is too high Heikal et al. (2014). The debt-to-equity ratio is calculated as:

$$D/E\text{ratio} = \frac{TotalLiabilities}{TotalStockholders'Equity}$$

- **Current Ratio:** The current ratio is a liquidity financial ratio that measures a company's ability to pay back short-term debt obligations using current assets Heikal et al. (2014). It is calculated as:

$$Current\text{ratio} = \frac{CurrentAssets}{CurrentLiabilities}$$

This ratio highlights how a company can use its available resources to satisfy short-term debts Arkan (2016)

- **Fixed Asset Turnover ratio:** The fixed asset turnover ratio is an asset management ratio that measures a company's ability to manage and use its fixed assets to generate sales. It is calculated as follows:

$$Total\ Asset\ Turnover\ Ratio = \frac{Sales}{Average\ Fixed\ Assets}$$

A high asset turnover ratio means that a company is efficient in generating sales. Arkan (2016)

- **Interest rates:** The interest rate is the price of money that banks charge for loans or the price of money invested in different business operations Jareno Cebrian and Negrut (2016)
- **Inflation rates:** The inflation rate is the rise of the price of consumer goods and services. The consumer price index which represents the average market basket of consumer goods, reflects the spending patterns of a population group and it is sometimes used to obtain the inflation rate Jareno Cebrian and Negrut (2016)

4.4 The model

4.4.1 Data Preprocessing

Data preprocessing involves a collection of techniques that can be performed on a dataset to ensure the effective training of machine learning algorithms and to achieve the goal of accurate results. In

general larger datasets are heterogeneous in nature and are compiled from different data sources, and as a consequence results in imperfect data Singh and Singh (2020). Data preprocessing includes a wide variety of techniques such as data transformation, data integration, and reduction techniques to mention the least. For the purpose of this research the following have been considered;

1. **Data Cleaning:** The first stage of data preprocessing involves error detection and error repairing. Error detection involves identifying missing values, duplicates, typos or outliers in the dataset Chu, Ilyas, Krishnan and Wang (2016). There are several different techniques that can be used to detect the above issues such as Rcode methods and data visualisation tools. The dataset for this research was obtained already cleaned with no missing values or anomalies. However, it is always good practise to check for these issues, therefore to identify missing values the following rcode will be used:

`is.na()`

and to detect any outliers in the dataset, the box plot method will be used.

2. **Data Transformation:** The data transformation step involves changing the data values, structure or format of the data to make it easier for the machine to interpret the data and to learn complex relationships. Singh and Singh (2020). This step enables the user the ability to adapt the data to the requirements of the problem. For example, gradient descent based algorithms such as neural networks require data to be scaled in order for gradient descent to converge quicker to its minima. As mentioned in the previous section, the dataset for this research consists of 11 different features that are measured on different scales. Therefore, to achieve the desired results, a scaling technique known as Normalisation will be used.

Normalisation or Min-Max normalisation is a scaling technique used to change the values of the dataset to range between 0 and 1 Ali, Faraj, Koya, Ali and Faraj (2014). The formula used for data normalisation is defined as;

$$X = \frac{X - X_{min}}{X_{max} - X_{min}}$$

where X_{max} and X_{min} denote the minimum and maximum values for each feature or column.

3. **Data Integration:** This step can be considered as part of the data transformation step as it involves merging data from multiple sources. As mentioned previously in section 4.1, this dataset is obtained from two different sources, where the technical indicators were obtained from yahoo finance and the fundamental indicators obtained from Borsdata. This step involves consolidating both datasets into one larger dataset for training and testing

4.4.2 Recurrent Neural Networks

Recurrent neural networks are a special type of ANN that use sequential data or time series data as inputs for sequence prediction problems. What distinguishes RNN's from other neural nets are the special hidden layers known as memory states which were developed to preserve sequential information from previous steps Bandara, Bergmeir and Smyl (2020). Unlike ANN's, recurrent neural networks have recurrent loops seen in figure 2, that are adapted to address the temporal order and

dependencies within sequential data Hewamalage, Bergmeir and Bandara (2021). In the past decade, RNN's have achieved outstanding results in prediction performance and mastering the ability to learn long-term dependencies. Some applications of RNN's include text generation, speech recognition and time series prediction Che, Purushotham, Cho, Sontag and Liu (2018).

Figure 2 Illustrates the architecture of a traditional RNN unfolded in time. This architecture is defined as a many-to-many RNN since there is an output for every input value. The input value at each time step t is denoted as x_t and the corresponding hidden state at each time step is denoted as h_t . The weight parameters denoted as U , W and V are the shared weights of the RNN. The weight matrix U denotes the strength of the connection between the input value and the hidden state. The weight matrix W corresponding to the weight of the connection between hidden states, and lastly, the weight matrix corresponding to the connection between the hidden state and the output is denoted as matrix V Bandara et al. (2020).

The special characteristic of the RNN is its ability to retain information from previous states, this

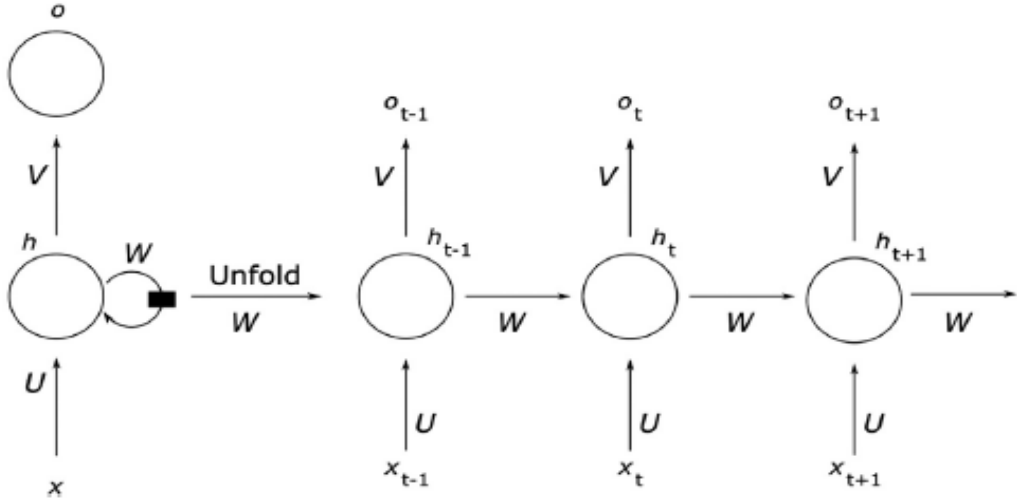


Figure 2. Figure 2: Recurrent Neural Network

retained information represents the memory of the model and is stored within each hidden state. Furthermore, the hidden states h_t are dependent on both the previous hidden state h_{t-1} and the current observation x_t . The hidden state h_t and the output o_t are defined as:

$$\begin{aligned} h_t &= f_{\theta}(Ux_t + Wh_{t-1}) \\ o_t &= f_{\alpha}(Vh_t) \end{aligned}$$

where o_t is the prediction at time step t and f_{θ} and f_{α} represent the non-linear activations functions such as the sigmoid, ReLu or tanh function Chen (2016). Once the predicted value has been generated, it is compared to the expected output value using a loss function. A loss function measures how different the predicted output is versus the expected output. There are different loss functions

that can be used to calculate prediction error, however, the choice of the loss function is dependent on the activation function used in the output layer. Generally for a regression model with a linear activation function in the output layer, the choice of loss function is the mean-squared error defined as;

$$\mathcal{L} = \frac{1}{N} \sum (\hat{y}_t - y_t)^2$$

Therefore, the goal of a training algorithm is to minimise the loss function based on the weights and bias terms using an optimisation process. This process is known as stochastic gradient descent where the algorithm searches the space of possible sets of weights for the model in order to improve predictions. The weights are then updated using Backpropagation for the next iteration. Since RNN's are time dependent, an extension of backpropagation is used known as backpropagation through time (BPTT) Bandara et al. (2020).

4.4.3 Backpropagation Through Time

The objective of Backpropagation is to update the weights and bias terms of the model in order to minimise the loss function and improve network performance. This algorithm involves computing the gradients of the loss function w.r.t. the weights of the network using partial derivations, we get the direction of decreasing loss, and during optimisation the weights move along this direction to eventually minimise the loss. This move along the negative direction is known as gradient descent and helps the network to learn and improve performance Lillicrap and Santoro (2019). Backpropagation Through Time differs from traditional Backpropagation in that the model is trained at a specific time t and at time steps that occurred before time step t such as $t-1$, $t-2$ and so on. This is accomplished by computing the gradients of the loss function with respect to the different weights of the network. This process can be expressed mathematically as follows:

let the loss function be $\mathcal{L} = \frac{1}{N} \sum (\hat{y}_t - y_t)^2$

and the predicted value at time step t be $\hat{y}_t$ which is computed in the output layer by the following equation:

$$\begin{aligned} \alpha_t &= (W_{ho} h_t + b_o) \\ \hat{y}_{t+1} &= \sigma(W_{ho} h_t + b_o) \end{aligned}$$

where σ denotes the activation function in the output layer, W_{ho} represents the weight of the connection between the hidden state and output layer, and b_o represents the bias term in the output layer.

Therefore, the minimisation of the loss function begins with taking the derivative of the $\mathcal{L}$ w.r.t. $\hat{y}_{t+1}$ and W_{ho} ;

$$\frac{\partial \mathcal{L}}{\partial \hat{y}_{t+1}} = (y_{t+1} - \hat{y}_{t+1})$$

Since weight (W_{ho}) is shared across all time steps, differentiation can occur at each step and sum all the results:

$$\frac{\partial \mathcal{L}}{\partial W_{ho}} = \sum_t^T \frac{\partial \mathcal{L}_{t+1}}{\partial \hat{y}_{t+1}} \frac{\partial \hat{y}_{t+1}}{\partial \alpha_{t+1}} \frac{\partial \alpha_{t+1}}{\partial W_{ho}}$$

Next is to calculate the gradient w.r.t. the bias term b_o

$$\frac{\partial \mathcal{L}}{\partial b_o} = \sum_t \frac{\partial \mathcal{L}}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial b_o}$$

To calculate the weight W_{hh} we should consider that it is shared across all time steps, therefore the gradient is calculated by taking into account all contributing time steps. For example, consider the gradient w.r.t. W_{hh} at time step t , that is, moving from $(t) \rightarrow t+1$, where only one time step is considered:

$$\frac{\partial \mathcal{L}_{t+1}}{\partial W_{hh}} = \frac{\partial \mathcal{L}_{t+1}}{\partial y_{t+1}} \frac{\partial y_{t+1}}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial W_{hh}}$$

However, since the hidden state at time step $t+1$ is dependent on the hidden state at t , backpropagation can be used to calculate the partial derivative;

$$\frac{\partial \mathcal{L}_{t+1}}{\partial W_{hh}} = \frac{\partial \mathcal{L}_{t+1}}{\partial y_{t+1}} \frac{\partial y_{t+1}}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t} \frac{\partial h_t}{\partial W_{hh}}$$

Therefore, by aggregating the gradients w.r.t W_{hh} over the whole time sequence, and further use backpropagation through time from $t+1$ to 1, the following gradient will be yielded:

$$\frac{\partial \mathcal{L}}{\partial W_{hh}} = \sum_t \sum_{k=1}^{t+1} \frac{\partial \mathcal{L}_{t+1}}{\partial y_{t+1}} \frac{\partial y_{t+1}}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_k} \frac{\partial h_k}{\partial W_{hh}}$$

Lastly, to derive the gradient at time step $t+1$ w.r.t. W_{xh} , we get the following:

$$\frac{\partial \mathcal{L}_{t+1}}{\partial W_{xh}} = \frac{\partial \mathcal{L}_{t+1}}{\partial y_{t+1}} \frac{\partial y_{t+1}}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial W_{xh}}$$

But because h_{t+1} depends on the contributions from h_t and X_{t+1} , this means to calculate the gradient of the loss function at time step t w.r.t W_{xh} , we need to backpropagate to h_t . Furthermore, if we sum up all the contribution from $t+1$ to 1 via backpropagation, and we take the derivative over the whole sequence we can yield the following gradient:

$$\frac{\partial \mathcal{L}}{\partial W_{xh}} = \sum_t \sum_{k=1}^{t+1} \frac{\partial \mathcal{L}_{t+1}}{\partial y_{t+1}} \frac{\partial y_{t+1}}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_k} \frac{\partial h_k}{\partial W_{xh}}$$

Chen (2016) The limitations of the backpropagation algorithm is that as the algorithm moves downwards from the output layer towards the input layer, the gradients get smaller and approaches zero which eventually leaves the weights of the initial or lower layers unchanged. This is known as the vanishing gradient and as the gradient approaches zero, the improvement calculated will be too small to make a difference to the initial weight. i.e. The vanishing gradient problem does not the algorithm to learn patterns from data that is more than 10 time steps behind the current value t Hanin (2018) To solve this problem, Hochreiter & Schmidhuber developed the long short term memory, which can learn longer time dependencies using special cells called memory states, which store relevant information which can be used in each step to help the algorithm learn better Yunpeng, Di, Junpeng and Yong (2017).

4.4.4 Long Short-term memory (LSTM)

The Long short-term memory algorithm is a unique type of recurrent neural network that was developed to solve the vanishing gradient problem Nikou et al. (2019). The difference between this algorithm and the RNN is that it was built to store information for longer time periods. Thus, allowing the algorithm to learn from past information. This is accomplished by the memory state of the LSTM cell, which stores all relevant information from previous cell states. Moreover, the flow of information is controlled by three special gates, mainly the forget gate, the input gate and the output gate. These three gates Nelson et al. (2017).

Architecture

Figure 3 shows a LSTM model with 3 repeated cells, where the number of cells correspond to the length of the forecasting window. For example suppose the forecasting window is 6, that is t_0 to t_6 then the LSTM will have 6 cells. Within each cell there is the cell state, forget gate, input gate and

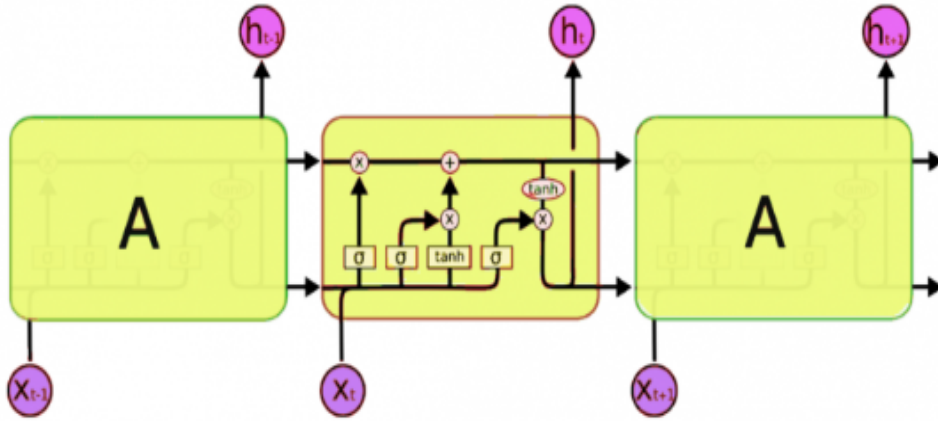


Figure 3. Full LSTM

an output gate, which act like neural networks to control the flow of information into and out of the cell, shown in figure 4 .

1. Cell state

The cell state C_t or the long-term memory of the network is a chain-like component that stores relevant information from each step and passes it on to the next step. The cell state is not responsible for altering any information, however it allows for the cell gates to delete or modify information through pointwise multiplication. Chen (2016).

2. Forget gate

The forget gate F_t is a neural network that takes as inputs x_t which corresponds to the inputs at time t and the hidden state from the previous time step h_{t-1} . This neural network determines which pieces of information from x_t and h_{t-1} are relevant to update the cell state at the current step Chen (2016). In order to filter this new information, the forget gate uses a sigmoid function

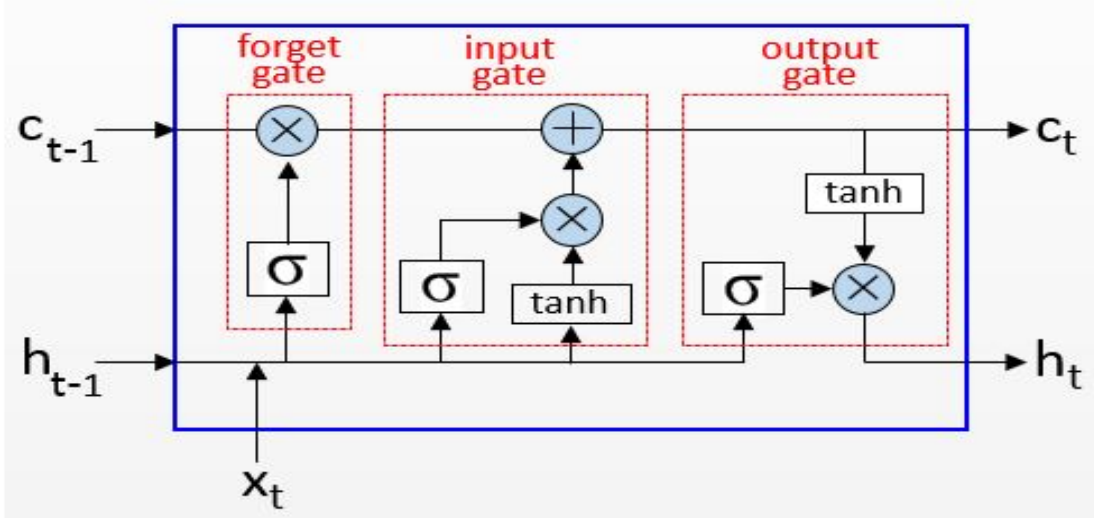


Figure 4. LSTM cell

which will return a matrix of values between 0 and 1 . This matrix is then pointwise multiplied to cell state C_t ,where information in the cell state that is deemed important will be multiplied by 1 ,whereas information that is deemed irrelevant will be multiplied by 0 . Thus information that is irrelevant will have little to no influence on the next step. The forget gate is defined by the following expression:

$$F_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$

where W_{xf}, W_{hf} are the weights associated with the connections in the forget gate, and b_f denotes the bias term Chen (2016).

3. Input gate

The main objective for this gate is to determine what new information should be added to the networks long-term memory C_t given the previous hidden state and the new input data. To update the cell state with new information, the input gate passes $x - t$ and h_{t-1} into two neural networks Moghar and Hamiche (2020). First, the input values at time step t and the hidden state from the previous time step are passed into a sigmoid function.

$$i_t = \sigma(W_{xi} + W_{hi}h_{t-1} + b_i)$$

The result of the sigmoid function will return a 0 for information that needs to be forgotten and 1 for information that needs to be retained. Next, x_t and h_t are passed into a tanh function which acts as a squashing function that returns a matrix of potential new candidates with values between -1 and 1.

$$g_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$$

Then , the results of i_t and g_t are pointwise multiplied to produce new candidates which are then pointwise added to the cell state to update the long-term memory Di Persio and Honchar (2017)

4. Output gate

Lastly, the output gate is responsible for determining the new hidden state and controls the flow of relevant information from one hidden cell to the next. This process occurs in two phases. First, x_t and h_{t-1} are passed into the third sigmoid function

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$

Then, the new cell state is passed through the tanh activation function to produce a matrix of values between -1 and 1. Lastly, o_t and $\tanh(c_t)$ are pointwise multiplied to produce the new hidden state;

$$h_t = o_t \cdot \tanh(c_t)$$

Therefore, the output of the cell will be a new cell state c_t and the new hidden state h_t which will be carried over to the next step Pang, Zhou, Wang, Lin and Chang (2020).

Typically in an LSTM, prediction occurs in the last LSTM layer. The output in the final layer are values between 0 and 1 which represent predicted probabilities.

4.5 LSTM Parameter Initialisations

4.5.1 LSTM Input layer

The first hidden layer of the LSTM network is the input layer. The input to every LSTM layer must be three-dimensional when fitting the model and making predictions. The three dimensions of this input are;

- **Samples:** One sequence is one sample. Therefore, A batch is comprised of one or more samples. Thus, in the context of this research to predict the next months stock prices, a look back period of 30 is needed, which results in the following batch size:

$$n = 2265 \text{ observations; lags} = 30; \text{batch size} = \frac{2265}{30} = 75$$

- **Time steps:** One time step is one point of observation in the sample. Since the goal is to predict the next 30-days of stock prices, a look back period of 30 is required.
- **Features:** Features correspond to the input variables or predictors of the output variable. For the purpose of this research, the total number features or predictors that will be used to predict the closing price is 11.

Therefore, the first input layer takes the input argument: [samples, timesteps , features].

4.5.2 Keras Model

In this research, R studio will be used to build the LSTM model and to forecast the closing price with respect to different indicators. The required packages used to run the model include Keras and tensorflow. The loss function and optimisation that will be used for training:

```
lstm_model %>%
compile(loss = 'mae', optimizer = 'adam', metrics = 'accuracy')
```

The main objective is to build a LSTM model with both fundamental indicators and technical indicators and to determine whether this model outperforms models that only considers one type of strategy. Therefore to achieve this objective, 4 different LSTM models will be built with the following layer specifications:

- Experiment 1: LSTM model with Fundamental Indicators

```
lstm_model <- keras_model_sequential()
lstm_model %>% layer_lstm(units = 50, # size of the layer
batch_input_shape = c(75, 30, 5), # batch size, timesteps, features
return_sequences = TRUE,
stateful = TRUE) %>%
# fraction of the units to drop for the linear transformation of the inputs
layer_dropout(rate = 0.5) %>%
layer_lstm(units = 50,
return_sequences = TRUE,
stateful = TRUE) %>%
layer_dropout(rate = 0.5) %>%
time_distributed(keras::layer_dense(units = 1)))}
```

- Experiment 2: LSTM model with Technical Indicators

```
lstm_model <- keras_model_sequential()
lstm_model %>% layer_lstm(units = 50, # size of the layer
batch_input_shape = c(75, 30, 6), # batch size, timesteps, features
return_sequences = TRUE,
stateful = TRUE) %>%
# fraction of the units to drop for the linear transformation of the inputs
layer_dropout(rate = 0.5) %>%
layer_lstm(units = 50,
return_sequences = TRUE,
stateful = TRUE) %>%
layer_dropout(rate = 0.5) %>%
time_distributed(keras::layer_dense(units = 1)))}
```

- Experiment 3: LSTM model with Fundamental Indicators and Technical Indicators

```

lstm_model <- keras_model_sequential()
lstm_model %>% layer_lstm(units = 50, # size of the layer
batch_input_shape = c(75, 30, 11), # batch size, timesteps, features
return_sequences = TRUE,
stateful = TRUE) %>%
# fraction of the units to drop for the linear transformation of the inputs
layer_dropout(rate = 0.5) %>%
layer_lstm(units = 50,
return_sequences = TRUE,
stateful = TRUE) %>%
layer_dropout(rate = 0.5) %>%
time_distributed(keras::layer_dense(units = 1))

```

4.6 Model Evaluation

The objective is to build a LSTM model with both fundamental indicators and technical indicators as inputs to predict the next months stock prices. Historical prices are used to train the model and for the algorithm to learn the relationships between the different input variables. Once the model is trained it will be used to predict the next months stock price. The focus of this section is on the evaluation of the performance of the different models, and how different combinations of input variables can affect the prediction accuracy of the model. Generally most machine learning problems can be broken down to either Regression or Classification problem, just like performance measures. Therefore, since the output of this model is a continuous variable, the focus for this section will be on regression metrics.

Evaluation metrics used for classification models are different than for regression models, since classification problems are mainly concerned with whether or not a prediction was correct or incorrect. However, for regression models we are predicting on a continuous range, thus having a different set of evaluation metrics Galdi and Tagliaferri (2018). The most commonly used regression metrics include; Mean absolute error (MAE), Mean Square Error (MSE), Root mean Squared Error (RMSE) and R^2 R-squared.

4.6.1 Mean Squared Error (MSE)

The mean square error metric is one of the most popular regression metric that has been used to measure the amount of error in a model. It finds the average of the squared difference between the target value and the value predicted by the model Engbrechtsen (2019). For models that predict continuous variables, the MSE is an ideal performance benchmark because of its link to the concept of cross-entropy. Since the goal is to identify the model that most closely produces the true stock price trend, then the best model will minimise the difference between predictions and the training data Hodson, Over and Foks (2021). The formula for MSE is represented as follows;

$$MSE = \frac{1}{N} \sum_{j=1}^N (y_j - \tilde{y}_j)^2$$

where $\tilde{y}_j$ denotes the predicted value, y_j denotes the actual value, and N denotes the number of points. Therefore, the closer the forecasts are to the actual values i.e. smaller the MSE, this indicates a better performing model and better fit.

5. Results

The purpose of this research was to build a deep learning algorithm that could predict future stock prices using both fundamental indicators and technical indicators. The chosen deep learning algorithm was the Long short-term memory algorithm, mainly for its memory capabilities and being able to handle sequential data. As described in the previous section, this model was going to be used under three different experiments. (Experiment 1) involved building a LSTM model with only technical indicators as inputs to predict the closing price, we shall call this the Technical model. (Experiment 2) involved predicting the closing price using fundamental indicators, the Fundamental model. Lastly, (Experiment 3) involved using both fundamental and technical indicators to predict the closing price, we shall call this the Hybrid Model. Due to the challenges faced in determining the results, this research will no longer be using the proposed Long short-term memory model as proposed in the previous sections. Instead, An Artificial neural network was used as an alternative model to the proposed LSTM. The challenges faced and reasons for adapting the research methodology, are outlined in the next section.

5.1 Research Challenges

- **R code:** The proposed model was to be built using R-language in R-studio. Keras() and tensorflow() R packages are two of the main of the main packages used to run an LSTM model in R. Keras() is an advanced API used for fast experimentation. Tensorflow() is an end-to-end, open-source machine learning platform. Both packages are written in Python, which means a python environment is needed to run the platforms. An attempt was made to use the Reticulate () package which allows for the use of Python code, scripts and modules with an R environment.
- **Python code:** By embedding a Python session within R, tensorflow and keras would run smoothly without interrupts. However, due to the lack of knowledge in python programming language, this solution would not work.
- **Insufficient material:** The research problem for this research is unique in that it attempts to combine two investment analysis techniques in a model that is generally used for text generation, video classifying and keyword spotting. Thus, using this model for stock price prediction is still a new topic being explored. The research papers that have used this model for stock price prediction, have used it as a univariate model, with only opening price as the input variable.

Therefore, the attempt to build a multiple input long-short term memory model was unsuccessful due to a lack of knowledge in Python language and the uniqueness of the this research problem which required more indepth information than there was currently available.

5.2 Solution

Despite not being able to use the long-short term memory model, the proposed strategy to build a model with both fundamental indicators and technical indicators will still be implemented using an Artificial Neural Network. This solution is achievable because it can be performed in Rstudio using R language and ANN's are much more simple and easier to implement on any problem.

5.2.1 *Technical Model*

The Technical model was built using the following input variables;

- Opening price, Volume, Highest price, Lowest price, Adjusted closing price
- Moving average, Exponential Moving average, Relative-strength Index

To build the best performing neural network, three different models were built using different hyper-parameter settings. The choice in the number of neurons in the hidden layer was based on the size of the input layer and the output layer **Model 1:**

1. Hidden layer: 2 hidden layers, one layer with 4 nodes and the second layer with 1 node (4,1)
2. activation function: Logistic
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. SSE=0.0079

Model 2:

1. Hidden layer: 2 hidden layers one with 4 nodes and the second with 1 node (4,1)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. SSE=0.0088

Model 3:

1. Hidden layer: 2 hidden layers one with 3 nodes and the second with 1 node (3,1)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. SSE=0.0077

Based on these results the best Technical model is Model (3), with 3 nodes in the first hidden layer and 1 nodes in the second hidden layer which can be seen in (Figure.5);

5.2.2 *Fundamental Model*

The Technical model was built using the following input variables;

- Opening price, Volume
- Interest rates, Inflation, Price/Book value, Price/Earnings, Earnings, Return on investment

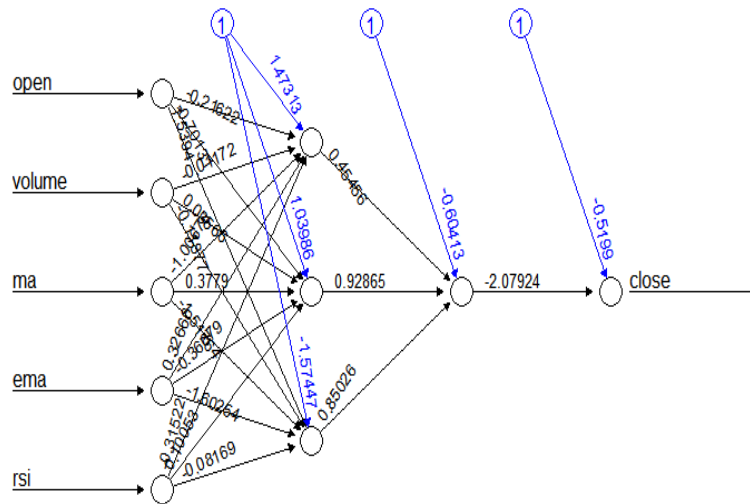


Figure 5. Technical Model

To build the best performing neural network, three different models were built using different hyper-parameter settings; **Model 1**:

1. Hidden layer: 2 hidden layers, one layer with 4 nodes and the second layer with 1 node(4,1)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. computed SSE=0.014

Model 2:

1. Hidden layer:2 hidden layers one with 3 nodes and the second with 2 node (3,2)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. Computed SSE=0.013

Model 3:

1. Hidden layer: 2 hidden layers one with 5 nodes and the second with 1 nodes (5,1)
2. activation function: Logistic
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. Computed SSE=0.0138

Based on these results the best Fundamental model is Model (2), with 3 neurons in the first hidden layer and 2 neurons in the second hidden layer, which can be seen in (Figure 6) below.

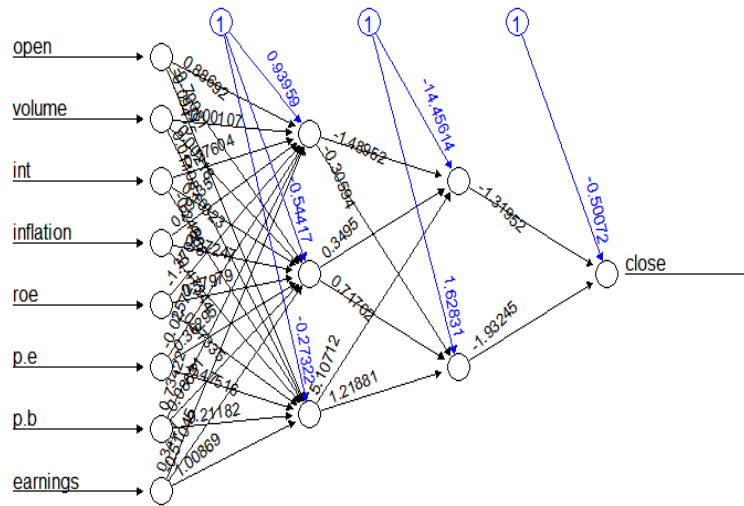


Figure 6. Fundamental Model

5.2.3 Hybrid Model

The Technical model was built using the following input variables;

- Opening price, Volume, Highest price, Lowest price, Adjusted closing price
- Moving average, Exponential Moving average, Relative-strength Index
- Interest rates, Inflation, Price/Book value, Price/Earnings, Earnings, Return on investment

To build the best performing neural network, three different models were built using different hyper-parameter settings;

Model 1:

1. Hidden layer: 2 hidden layers one with 4 nodes and the second with 1 node (4,1)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. computed SSE=0.0082

Model 2:

1. Hidden layer: 2 hidden layers one with 3 nodes and the second with 2 node (3,2)
2. activation function: tanh
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. Computed SSE=0.011

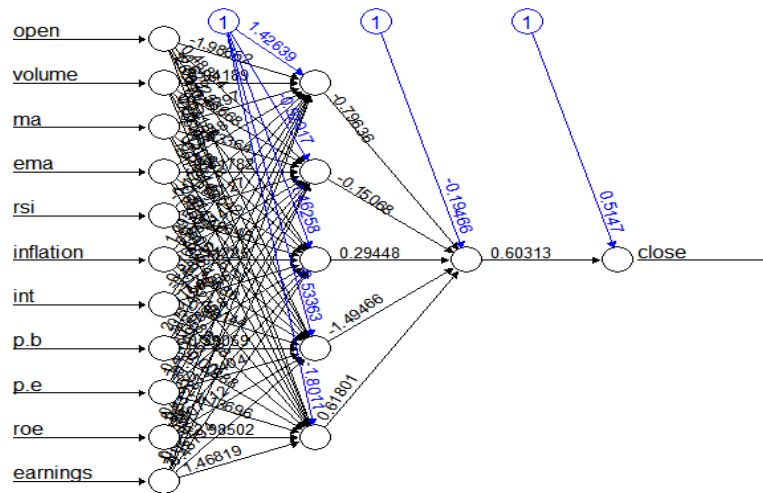


Figure 7. Hybrid Model

Model 3:

1. Hidden layer: 2 hidden layers one with 5 nodes and the second with 1 node (5,1)
2. activation function: Logistic
3. algorithm used for learning: rprop+ (resilient backpropagation)
4. Computed SSE=0.006

Based on these results the best Hybrid model is Model (3), with 5 neurons in the first hidden layer and 1 neurons in the second hidden layer, which can be seen in (Figure 7) below.

6. Conclusion

According to the results of this analysis it can be concluded that the artificial neural network with both fundamental indicators and technical indicators is the best performing model with a computed SSE of 0.0066. In second place was the Technical model with a SSE of 0.007 and lastly the Fundamental model produced a SSE of 0.013. Initially this study aimed to build a deep learning algorithm specifically the long short-term memory model to predict stock prices, however, due to the research challenges mentioned in the paper, an alternative approach was used. An artificial neural network is a model that has gained immense popularity in finance and trading. It has the ability to recognise patterns in complex data and the flexibility to adapt to changing variables. Despite not using the LSTM model, the neural network performed well in learning past movements of stock prices. Therefore, this research was successful in building on the idea that using both fundamental indicators and technical indicators as a hybrid strategy results in a better performing model. Furthermore, by using both investment strategies an investor has the opportunity to view the market from different perspectives, and thus gains a holistic view of the market and the company.

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